Sri Lanka Institute of Advanced Technological Education




Higher National Diploma in Information Technology

Advanced Technological institute, Dehiwala

Individual project proposal

# Smart Study Planner with AI Suggestions

Student name : P.A.Tharukshi Hansamali

Registration number: DEH/IT/2023/F/0068

## Statement of Originality

I hereby declare that this project proposal is my own original work and has been developed based on my own ideas and understanding. This work has not been copied or submitted from any other source, either in whole or in part.

All the information included in this proposal has been carefully studied and organized by me. Any external sources, references, or ideas that have been used for guidance have been properly understood and are not directly copied.

I confirm that this proposal is prepared independently as part of my academic requirement, and I take full responsibility for the content presented in this document.

Signed,

P.A.Tharukshi Hansamali

DEH/IT/2324/F/0068

07/04/2026

## Table of Content


1. Introduction 4
2. Problem Statement 4
3. Objectives 5
4. Scope of the System 5
5. Proposed Solution 6
6. Methodology 6
7. Technologies to be Used 6
8. Expected Outcomes 7
9. Target Users 7
10. Time Line 7

# Smart Study Planner with AI Suggestions

## 1.Introduction

In today's education system, students face many challenges in managing their studies effectively. They have to handle multiple subjects, assignments, and exams within limited time periods. Due to poor time management and lack of proper planning, many students fail to achieve their academic goals.

Most students still rely on manual methods such as notebooks or basic planners to organize their studies. However, these methods are not flexible and cannot adapt to changing schedules or individual learning needs. As a result, students often neglect difficult subjects, miss important deadlines, and experience stress during exams.

With the advancement of technology, there is an opportunity to improve this situation using smart systems. This project proposes a **Smart Study Planner with AI Suggestions**, which helps students create personalized study plans and receive intelligent guidance. The system will analyze user input and provide recommendations to improve study efficiency and performance.

## 2.Problem Statement

- **Who faces the problem:**
  Students in schools, colleges, and universities who have multiple subjects, exams, and limited study time.
- **What the problem is:**
  Students find it difficult to create effective study plans manually. They may forget important tasks, fail to prioritize difficult subjects, and struggle to track their progress.
- **Why it needs to be solved:**
  An intelligent system can automate study planning, ensure better time management, and guide students to focus on important areas. This will improve learning outcomes and reduce academic stress.

# 3.Objectives

General Objective:

- ✓ To develop an intelligent system that automatically generates personalized study plans and provides AI-based suggestions to optimize learning efficiency.

Specific Objectives:

- ✓ To allow users to input subjects, topics, difficulty levels, deadlines, and available study time
- ✓ To automatically generate daily and weekly study timetables
- ✓ To prioritize difficult subjects and urgent deadlines
- ✓ To provide reminders and notifications for pending tasks
- ✓ To analyze student performance and identify weak areas
- ✓ To provide AI-based suggestions for improving study habits
- ✓ To design a user-friendly and interactive interface

# 4.Scope of the System

In Scope:

- ✓ Web-based application accessible on desktop and mobile browsers.
- ✓ Automatic generation of personalized daily and weekly study timetables.
- ✓ AI-based recommendations for weak subjects, missed tasks, and exam preparation.
  Progress tracking and visualization (graphs and percentage completion).
- ✓ Notifications and reminders for pending tasks.

Out of Scope:

- ✓ Offline desktop-only applications.
- ✓ Direct integration with school learning management systems (LMS).
- ✓ Advanced AI predictions using Machine Learning models (beyond rule-based AI).

# 5.Proposed Solution

- ❖ Type of System: - Web application
- ❖ Key Features:-
  - ✓ User registration and profile management.
  - ✓ Subject and task management.
  - ✓ AI-powered smart timetable generation.
  - ✓ Intelligent suggestions and notifications.
  - ✓ Progress tracking dashboard.
  - ✓ Notification and reminders.

- ❖ How it solves the problem:-

  The system reduces the burden of manual study planning by automatically generating schedules based on user input. It helps students focus on weak subjects and upcoming deadlines. If a student misses a task, the system dynamically adjusts the schedule. AI-based suggestions guide students to improve their study habits and performance.

# 6.Methodology

- The system will be developed using Agile methodology, with iterative development and testing.
- Each feature (UI, backend, AI logic) will be implemented and tested in small modules before integration
- User feedback will be incorporated to refine AI suggestions and timetable generation.

# 7.Technologies to be Used

- Frontend: React.js
- Backend: Node.js
- Database: MongoDB
- Tools: Visual Studio Code
- AI Logic: OpenAI API or rule-based AI logic for intelligent suggestions

## 8.Expected Outcomes

- Improved study planning and time management
- Reduced errors in manual scheduling
- Better focus on important subjects and deadlines
- Personalized study recommendations
- Enhanced user experience through interactive dashboards

## 9.Target Users

- Students preparing for exams
- School and university students
- Educational institutions

## 10.Timeline

| Tasks | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Project proposal submission & approval | | | | | | | | | | | | |
| SRS submission | | | | | | | | | | | | |
| UI/UX design & database design | | | | | | | | | | | | |
| Frontend & backend development | | | | | | | | | | | | |
| AI logic implementation | | | | | | | | | | | | |
| Progress tracking, notifications, testing | | | | | | | | | | | | |
| Final testing & report preparation | | | | | | | | | | | | |